

PATIENT DETAILS

Ms. Divya Raghunathan

28 Years | Female | Unmarried

Blood Group: B+ | Weight: 52 kg | BMI: 19.8

Chief Complaint: Palpitations, Fatigue, Dyspnoea on exertion (3 months)

MR Number

MH-BLR-2024-38841

Referring Doctor

Dr. Sujatha Krishnan, MD – Cardiology

Department

Cardiology OPD

Sample Type

Venous Blood + Spot Urine

Collection

06 Jun 2026 08:00 AM (Fasting)

Report

06 Jun 2026 01:00 PM

■ COMPLETE BLOOD COUNT (CBC)

Panel: CBC-001 | Haematology

TEST	RESULT	UNIT	NORMAL RANGE	FLAG
Haemoglobin (Hb)	7.4	g/dL	12.0 – 16.0 (F)	CRITICAL
RBC Count	3.1	x10 ¹² /uL	3.8 – 5.2 (F)	LOW ▼
Haematocrit (PCV)	22	%	35.0 – 47.0 (F)	LOW ▼
MCV	62	fL	80 – 100	LOW ▼
MCH	18	pg	27 – 33	LOW ▼
MCHC	29	g/dL	31.5 – 36.0	LOW ▼
RDW-CV	18.5	%	11.5 – 14.5	HIGH ▲
WBC Count	9.8	x10 ³ /uL	4.5 – 11.0	Normal
Platelets	182	x10 ³ /uL	150 – 400	Normal
Reticulocyte Count	0.6	%	0.5 – 1.5	Normal

■ IRON STUDIES & HAEMATINIC PROFILE

Panel: IRN-002

TEST	RESULT	UNIT	NORMAL RANGE	FLAG
Serum Iron	28	ug/dL	60 – 170	LOW ▼
Serum Ferritin	6	ng/mL	12 – 150 (F)	LOW ▼
TIBC (Total Iron Binding Cap)	490	ug/dL	250 – 370	HIGH ▲
Transferrin Saturation	5.7	%	20 – 50	LOW ▼
Serum Transferrin	415	mg/dL	204 – 360	HIGH ▲
Soluble Transferrin Receptor	6.2	mg/L	0.8 – 2.8	HIGH ▲
Vitamin B12	142	pg/mL	200 – 900	LOW ▼
Folic Acid (RBC Folate)	3.2	ng/mL	3.0 – 17.0	Normal
Vitamin D (25-OH)	11	ng/mL	30 – 100	LOW ▼

CARDIAC BIOMARKERS

Panel: CARD-003 | STAT

TEST	RESULT	UNIT	NORMAL RANGE	FLAG
Troponin I (hs-cTnI)	0.028	ng/mL	< 0.040	Normal
BNP (Brain Natriuretic Peptide)	285	pg/mL	< 100	HIGH ▲
NT-proBNP	620	pg/mL	< 125 (< 50 yrs)	HIGH ▲
CK-MB	18	U/L	< 25	Normal
hs-CRP	4.8	mg/L	< 1.0	HIGH ▲
Homocysteine	22.1	umol/L	5 – 15	HIGH ▲
Ferritin (Cardiac context)	6	ng/mL	12 – 150	LOW ▼

THYROID FUNCTION

Panel: THY-004

TEST	RESULT	UNIT	NORMAL RANGE	FLAG
TSH	9.12	uIU/mL	0.27 – 4.20	HIGH ▲
Free T4 (fT4)	0.68	ng/dL	0.93 – 1.70	LOW ▼
Free T3 (fT3)	2.1	pg/mL	2.0 – 4.4	Normal
Anti-TPO Antibodies	245	IU/mL	< 35	HIGH ▲

INTEGRATED CLINICAL IMPRESSION

Primary Diagnosis: Severe Iron Deficiency Anaemia (IDA) with Cardiac Strain + Hypothyroidism

- Severe IDA** — Hb 7.4 g/dL, MCV 62, MCH 18, Ferritin 6 ng/mL, TSAT 5.7%. Microcytic hypochromic picture classic for iron deficiency. IV Iron Sucrose (200 mg x 5 doses) recommended given severe depletion and GI intolerance risk.
- Anaemia-Induced Cardiac Strain** — Elevated BNP (285) and NT-proBNP (620) without hs-Troponin rise suggests high-output state from anaemia. Echocardiography advised; symptoms likely to resolve with Hb correction. Holter monitoring for palpitations.
- Hypothyroidism (Hashimoto's)** — TSH 9.12, fT4 0.68, Anti-TPO 245 IU/mL. Initiate Levothyroxine 50 mcg once daily; recheck TFT in 6–8 weeks. Thyroid dysfunction exacerbates anaemia and cardiac symptoms.
- Nutritional Deficiencies** — Vit D 11 ng/mL critically low; B12 142 pg/mL deficient. Supplement: Cholecalciferol 60,000 IU weekly x 8 wks + Methylcobalamin 1500 mcg/day.
- Follow-up** — Repeat CBC, Iron, TFT, BNP at 6 weeks. GI endoscopy to identify source of chronic blood loss (menorrhagia? GI bleed?). Gynaecology referral advised.

Dr. Sujatha Krishnan, MD DM Consultant –
Cardiology Manipals Hospitals Bangalore MCI
Reg: 72481

Dr. Roopa Nair, MD Consultant – Clinical Pathology
Manipals Lab Services MCI Reg: 55903

06 Jun 2026 | 01:00 PM LIS:
Manipal LabOS v6.1 Digitally
Validated Report Status: FINAL